How Far Can Structure-Based Machine Learning Predict Experimental Singlet–Triplet Gaps? An Honest Benchmark for TADF Emitter Triage

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Discovering thermally activated delayed fluorescence (TADF) emitters hinges on the singlet–triplet energy gap (ΔE_{ST}), yet high-level excited-state calculations are costly and disagree with experiment by several tenths of an electronvolt. We ask a deliberately narrow question: how accurately can machine learning estimate *experimental* ΔE_{ST} from the molecular structure alone, and what limits it? Regressing ΔE_{ST} on features that include the S_1 and T_1 energies is circular—since $\Delta E_{\mathrm{ST}} \equiv E_{S_1} - E_{T_1}$, the model merely subtracts two inputs—so we target the measured gap instead. Using 231 donor–acceptor molecules with literature-reported gaps, we train random forests on three descriptor sets and evaluate them under Bemis–Murcko scaffold cross-validation (212 distinct scaffolds), holding whole scaffolds out of training. Models built from the 2D structure alone—Morgan fingerprints (MAE 0.091 eV) and RDKit descriptors (0.100 eV)—match descriptors derived from a semi-empirical excited-state calculation (0.096 eV); the excited-state step adds nothing. All sets reach $R^2 \approx 0.25$–0.31 with wide confidence intervals and weak ranking (Spearman 0.26 to 0.36); the signal is modest but real—below a label-permutation null at $p = 0.001$—and only slightly above a mean-value baseline (MAE 0.107 eV). Accuracy is bounded near 0.1 eV by two effects we quantify: scatter among independent experimental reports (0.1 eV to 0.15 eV) and functional dependence of the computed reference (up to 0.58 eV). As a triage filter for low-gap emitters the model enriches only modestly (1.2 to 1.4× over random). The practical message is cautionary: accuracy is set by data quality, not model capacity. All data, models, and code are openly available (Zenodo DOI: 10.5281/zenodo.17436069).

1 Introduction

1.1 Application-driven materials discovery: horticultural and human-centric lighting

Controlled-environment agriculture and human-centric lighting both demand spectrally tunable, energy-efficient light sources: lighting can reach 40–60% of operating costs in vertical farms[1], and matching emission to chlorophyll absorption or circadian (melanopic) response bands improves efficiency[2,3]. Commercial inorganic LEDs rely on rare-earth phosphors (Eu, Tb, Y) with broad emission and supply-chain risk[4], motivating tunable, heavy-metal-free organic alternatives.

Thermally activated delayed fluorescence (TADF) materials are such an alternative: organic emitters that harvest both singlet and triplet excitons through donor–acceptor architectures, achieving near-unity internal quantum efficiency *without* heavy metals[5]. Their emission is molecularly tunable and their synthesis avoids rare-earth supply-chain risks[6,7]. However, realizing this potential requires navigating a vast chemical space of donor–acceptor combinations while satisfying competing requirements: a minimal singlet–triplet energy gap ($\Delta E_{\mathrm{ST}} < 0.2\,\mathrm{eV}$); efficient reverse intersystem crossing ($k_{\mathrm{RISC}} > 1 \times 10^6\,\mathrm{s}^{-1}$); and spectral overlap with target photoreceptor absorption bands. This paper addresses only the first of these—the ΔE_{ST} gap. The rate-constant and device-efficiency requirements demand spin–orbit-coupling and device-level calculations beyond the cheap structural data studied here, and we make no quantitative claim about them (section 3.8).

1.2 The computational bottleneck and the role of machine learning

Designing such emitters means searching a large space of donor–acceptor combinations for a small ΔE_{ST}. The direct route—time-dependent DFT on excited states—scales steeply with system size[8], so exhaustive screening of more than a few hundred candidates is impractical. Machine learning offers a cheaper surrogate[9,10], and several groups have predicted TADF gaps from molecular descriptors[11,12]. Those low reported errors, however, are obtained against *computed* (TD-DFT or higher) reference gaps; predicting the *experimental* gap is markedly harder, to the point that one study abandoned direct regression for classification[13] (section 3.7).

Two problems recur in this literature, and we confront both. First, accuracy is often quoted against a single random train–test split, which rewards interpolation within familiar scaffolds and overstates how a model behaves on new chemistry. Second, and more subtle, models are sometimes trained to reproduce a *computed* gap from a feature set that already contains the constituent excitation energies; because $\Delta E_{\mathrm{ST}} \equiv E_{S_1} - E_{T_1}$, this reduces to subtraction and the reported accuracy reflects arithmetic rather than learning.

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Our earlier work set the stage: Article 1[14] benchmarked GFN2-xTB/sTDA against 747 experimental molecules (MAE = 0.162 eV), and Article 2[15] extracted structure–property design rules from that set. Here we ask the narrower, sharper question those studies left open: starting only from cheap structural information, how accurately can a model predict the *measured* ΔE_{ST}, and what sets the limit?

1.3 Our contributions: an honest benchmark of structure-based prediction

Rather than claim a new high-accuracy predictor, we ask how far cheap, structure-based machine learning can go in predicting the *experimental* singlet–triplet gap—and answer carefully, explicit about where it fails. The primary result is a scaffold-validated benchmark with a calibrated accuracy ceiling: 2D structure alone predicts the measured gap as well as semi-empirical excited-state features, and we quantify why neither does better. A secondary, cautionary contribution is methodological—a reminder that excited-state-energy *leakage* inflates reported accuracy for any property defined as a difference of computed quantities. We make four contributions.

Structure alone, with no excited-state calculation, does as well as anything. On 231 molecules across 212 Bemis–Murcko scaffolds, random forests built from the 2D structure alone—Morgan fingerprints (MAE = 0.091 eV) and RDKit descriptors (0.100 eV)—match descriptors derived from a semi-empirical excited-state calculation (NTO spatial features, 0.096 eV). The result holds across scaffold, cluster and random cross-validation and against a label-permutation null, so it is not a splitting artefact. The excited-state step buys nothing for the S_1–T_1 gap; SHAP nonetheless ties the NTO variant to hole–electron overlap and charge transfer, giving an interpretable read on what structure encodes.

An explicit, tested accuracy ceiling. We quantify why accuracy saturates near 0.1 eV: independent experimental reports for the same compound disagree by 0.1 eV to 0.15 eV, and the computed reference is functional-dependent (B3LYP vs CAM-B3LYP differ by up to 0.58 eV). The ceiling is not an excuse: no higher-capacity learner (gradient boosting, MLP, SVR) beats the random forest, and the result places itself honestly at $R^2 \approx 0.25$, a real but limited signal bounded by data quality rather than model capacity.

A cautionary note on feature–target leakage. Secondary to the benchmark, we flag that regressing ΔE_{ST} on features containing the S_1/T_1 energies is circular ($\Delta E_{\mathrm{ST}} \equiv E_{S_1} - E_{T_1}$): the model returns arithmetic, invisible to cross-validation. Such feature–target leakage is a recognised cause of over-optimistic ML results[16] and is singled out in best-practice guidance for chemistry,[17] yet remains easy to introduce whenever the target is a derived energy difference.

Modest, quantified triage utility—and no device claims. As a pre-screen for TADF viability, the model enriches low-gap candidates (1.2 to 1.4× over random selection), and we deposit a conformal-bounded ranked shortlist of the enumerated library—useful for prioritisation but far from decisive. ΔE_{ST} is necessary but not sufficient for TADF efficiency; emission-wavelength matching for any specific lighting niche is a separate, downstream step not addressed here. We make no device-level performance claims; our earlier spin–orbit-coupling estimates of k_{RISC} and external quantum efficiency proved molecule-independent and were withdrawn.

This study builds on our earlier work—**Article 1**[14] (xTB/sTDA validation) and **Article 2**[15] (design rules)—by asking, rigorously, what such cheap data can and cannot support. All data, trained models, and code are openly available (Zenodo DOI: 10.5281/zenodo.17436069), enabling reproduction and extension of the benchmark.

2 Methods

2.1 Experimental target and dataset

The prediction target is the *experimental* singlet–triplet gap ΔE_{ST}. We extracted literature-reported values from a curated corpus of TADF publications and matched them, by compound name, to molecules in our descriptor set, retaining values in the physical range −0.3 eV to 1.5 eV. Where a compound carried multiple independent reports we used the median and recorded the inter-report spread; this scatter is typically 0.1 eV to 0.15 eV and bounds achievable accuracy (section 3.4). The final benchmark comprises 231 donor–acceptor molecules spanning 212 distinct Bemis–Murcko scaffolds[18].

The corpus does not record measurement medium (solvent, host) for individual values, so the per-molecule median is a consensus over uncontrolled conditions rather than a single, solvent-specific state. Because environmental electrostatics and conformational disorder are recognised as primary obstacles to accurate TADF modelling,[19] this consensus is best read as the central tendency of experimentally accessible S_1–T_1 gaps; it trades physical specificity for reduced variance. For the 102 molecules with repeat reports, a single random report deviates from this median by 0.072 eV on average—comparable to the model error itself (table S3)—which sets a floor on attainable accuracy independent of model capacity.

We deliberately do *not* train on the semi-empirical gap $E_{S_1} - E_{T_1}$: because this quantity is, by construction, a difference of two input features, regressing onto it measures arithmetic rather than learning (section 3.1).

2.2 Cheap electronic-structure inputs and physics-informed descriptors

Excited-state quantities used to build descriptors were obtained at the semi-empirical level: GFN2-xTB[20] geometries (`xtb` 6.5.1) followed by sTDA[21] for the lowest singlet and triplet states. From the natural transition orbitals (NTOs)[22] of each excitation we computed spatial overlap descriptors that encode exchange-integral physics:

$$S_{\mathrm{he}} = \int \phi_{\mathrm{hole}}(\mathbf{r})\, \phi_{\mathrm{electron}}(\mathbf{r})\, d\mathbf{r}, \tag{1}$$

$$\Delta\lambda_{\mathrm{A}} = |\lambda_{\mathrm{A}}(S_1) - \lambda_{\mathrm{A}}(T_1)|, \tag{2}$$

$$\mathrm{Char}^2_{\mathrm{diff}} = \Big[\textstyle\sum_i |q_i(S_1) - q_i(T_1)|\Big]^2, \tag{3}$$

$$\Delta S_{\mathrm{he}} = |S_{\mathrm{he}}(S_1) - S_{\mathrm{he}}(T_1)|. \tag{4}$$

S_{he} quantifies hole–electron overlap (related to the exchange integral), $\Delta\lambda_{\mathrm{A}}$ acceptor-participation differences, $\mathrm{Char}^2_{\mathrm{diff}}$ charge-transfer character changes, and ΔS_{he} the singlet–triplet overlap difference. The full set of 35 NTO descriptors (the S_1/T_1 energy scalars excluded) is listed in the Supporting Information; these descriptors require the sTDA excited-state calculation, even though they omit the energies themselves.

2.3 Feature sets and machine-learning models

We compared three feature representations on the same molecules. Two require no quantum chemistry, being computed from the 2D structure alone: (i) RDKit physicochemical descriptors[23] and (ii) Morgan fingerprints (radius 2, 2 048 bits). The third, (iii) the NTO spatial descriptors above, is derived from the sTDA excited state but excludes the energy scalars. A reference set that additionally includes the sTDA S_1/T_1 energies was also evaluated. We used random-forest regressors[24] (400 trees, scikit-learn[25]), which outperformed support-vector regression, ridge regression and a mean-value baseline on this task.

Evaluation. All performance figures use Bemis–Murcko scaffold `GroupKFold` ($k = 5$): every test molecule lies on a scaffold unseen in training, so reported errors estimate generalisation to new chemotypes rather than interpolation. We benchmark against a mean-value predictor and the raw sTDA gap, and report 95 % confidence intervals from 2 000 bootstrap resamples of the out-of-fold predictions.

Interpretability and reproducibility. Feature importance was quantified with exact TreeSHAP[26] on the fitted forest. All scripts (`code/finalize_model.py`, `code/rebuild_structural_model.py`) regenerate every reported number and fig. 1.

Bootstrap confidence intervals and power analysis. Point estimates of the rank correlation are complemented by 5 000-replicate percentile bootstrap confidence intervals on the out-of-fold predictions (`code/r2_ci_power_analysis.py`). For the headline model the Spearman $\rho = 0.36$ ($n = 231$) has a 95% CI of 0.24 to 0.47 ($p = 2.6 \times 10^{-8}$) and the Pearson $r = 0.50$ a 95% CI of 0.26 to 0.68; both exclude zero, establishing statistically significant association despite the wide R^2 interval. A power analysis for the correlation test (Fisher z transform, two-sided, $\alpha = 0.05$) shows that detecting $\rho = 0.36$ at 80 % power requires only $n \approx 60$ (90 % power: $n \approx 79$); at $n = 231$ the achieved power is ≈ 1.0. The ranking-ability question is thus amply powered, and the wide R^2 interval is *not* a sample-size problem: it reflects the high variance of a variance-explained statistic when a handful of high-gap outliers dominate the residual sum of squares. We therefore report Spearman ρ as the primary metric for triage utility, with the full set of metrics and intervals collected in table S5.

2.4 Reference-level benchmark

To bound the meaning of any model accuracy, we compared the computed reference across functionals for 7 representative emitters in gas and toluene, contrasting B3LYP/6-31G(d) with CAM-B3LYP/def2-TZVP (Gaussian 16[27]; ORCA 6.1.0[28]). The two levels differ in ΔE_{ST} by a mean absolute 0.30 eV (up to 0.58 eV, with cases of reversed relative magnitude); full values are tabulated in table S9. This functional dependence, together with experimental inter-report scatter, sets the accuracy ceiling discussed in section 3.4. The same CAM-B3LYP/def2-TZVP calculations additionally yielded excited-state-optimised (adiabatic) gaps and the two lowest triplets (T_1, T_2) for 14 emitters, used to bound the vertical approximation (section 3.4) and the triplet-manifold scope (section 3.8); see SI.

2.5 Heuristic enumeration of an extended library

To explore chemical space beyond the annotated set, we enumerated 22 194 donor–acceptor combinations from literature-validated fragments (carbazole, phenoxazine, acridine donors; triazine, benzonitrile, sulfone acceptors) under simple constraints (molecular weight $< 900\,\mathrm{Da}$, ≥ 2 aromatic rings, ≥ 1 N/O/S heteroatom). These were ranked by a rule-based TADF score

$$\text{TADF score} = 0.35\, f_{\mathrm{MW}} + 0.25\, f_{\mathrm{LogP}} + 0.20\, f_{\mathrm{rings}} + 0.20\, f_{\mathrm{hetero}}, \tag{5}$$

with each $f_i \in [0, 1]$ a normalised feature score based on established design heuristics[5,29,30]. The score is a coarse synthesis-planning filter, *not* a calibrated ΔE_{ST} predictor; the enumerated molecules are not quantum-validated and are reported as leads only.

2.6 Data and code availability

The experimental corpus, descriptor tables, trained models, and all analysis and figure-generation scripts are openly available on Zenodo (DOI: 10.5281/zenodo.17436069) under a CC-BY-4.0 licence, so that every number and figure in this work can be reproduced and the benchmark extended.

3 Results and Discussion

3.1 Why we target the measured gap: ruling out a circular benchmark

A natural temptation is to regress the singlet–triplet gap ΔE_{ST} onto a feature set that already contains the S_1 and T_1 excitation energies. Because $\Delta E_{\mathrm{ST}} \equiv E_{S_1} - E_{T_1}$ by definition, such a model reproduces its target to within numerical noise: in our data the sTDA gap equals $E_{S_1} - E_{T_1}$ with correlation 1.000 and zero mean absolute error. A support-vector regressor trained this way reaches an in-sample cross-validated $R^2 = 0.87$, but this number measures arithmetic, not learning—a direct subtraction of two input features outperforms the regressor. We therefore discard the semi-empirical gap as a target and benchmark models on the only non-trivial quantity: the *experimentally measured* ΔE_{ST}.

3.2 Structure-based models predict experimental gaps with modest, honestly-bounded accuracy

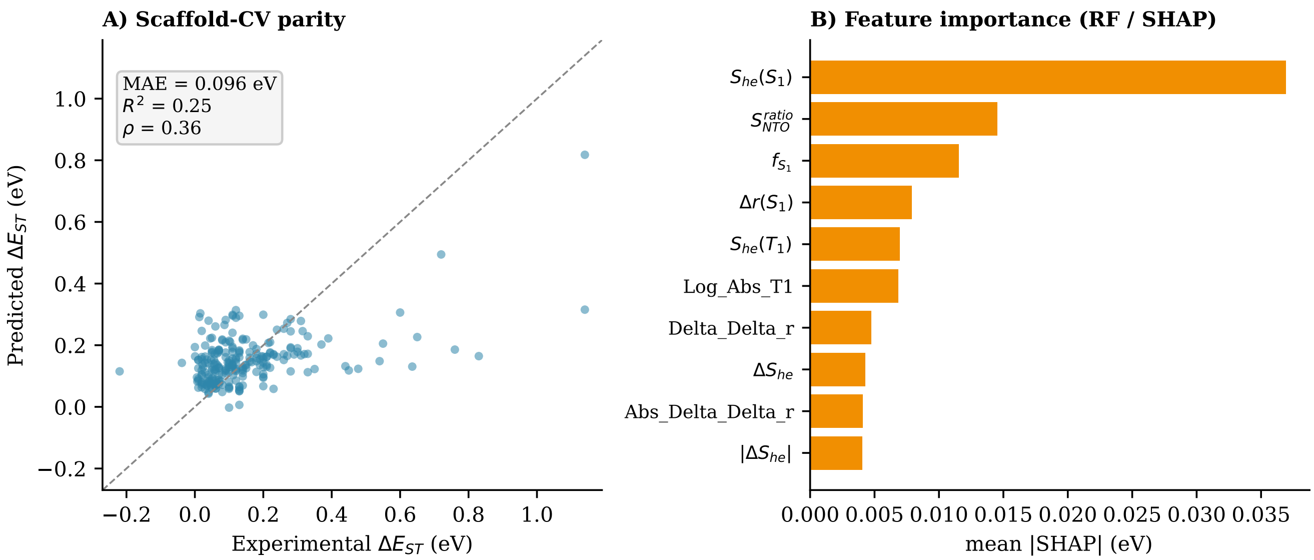

Figure 1 Structure-based prediction of experimental ΔE_{ST}. (A) Scaffold cross-validated parity for the random forest on NTO spatial descriptors (231 molecules, 212 Bemis–Murcko scaffolds): MAE = 0.096 eV, $R^2 = 0.25$, Spearman $\rho = 0.36$. Models built from the 2D structure alone (Morgan, RDKit) perform equivalently (table 1). The model regresses high-gap outliers toward the mean, the expected behaviour when the training distribution is dominated by low-gap emitters. (B) SHAP importances: hole–electron overlap S_{he}, NTO overlap ratio, oscillator strength f_{S_1} and charge-transfer distance Δr dominate, consistent with exchange-integral physics.

We assembled 231 donor–acceptor molecules for which an experimental ΔE_{ST} could be matched to a structure, and evaluated random-forest regressors under Bemis–Murcko scaffold cross-validation (212 distinct scaffolds), so that no scaffold appears in both training and test. Because 92.9 % of scaffolds are singletons, scaffold and random splits produce numerically similar folds (test–train nearest-neighbour Tanimoto differs by only 0.022). The protection is therefore chemical: zero shared scaffolds per fold, versus 3 to 8 under a random split. Task difficulty does not change substantially. That the model learns real signal rather than scaffold-group structure is confirmed by a label-permutation test: the true scaffold-CV MAE (0.096 eV) lies far below the null from 1 000 target shuffles (mean 0.121 eV, $p = 0.001$; table S8). The headline accuracy is also robust to the validation scheme: replacing the singleton scaffolds with genuine multi-molecule Butina clusters (down to 70 groups, largest 38 molecules) leaves the MAE at 0.099 eV to 0.104 eV, between scaffold-CV (0.096 eV) and random-CV (0.106 eV) (table S15). Three feature sets give statistically indistinguishable performance (table 1). Two are computed from the 2D structure alone, with no quantum chemistry: Morgan fingerprints (MAE = 0.091 eV) and RDKit physicochemical descriptors (0.100 eV). The third, NTO spatial descriptors, is derived from a semi-empirical sTDA excited-state calculation but excludes the S_1/T_1 energy scalars; it reaches MAE = 0.096 eV (95 % CI 0.082 to 0.109), $R^2 = 0.25$ (CI 0.03 to 0.45), Spearman $\rho = 0.36$. That the structure-only sets match the NTO set is the central methodological point: the excited-state calculation buys nothing, and adding back the S_1/T_1 energies does not help either (table 1). The honest comparator is the mean-value baseline (MAE = 0.107 eV), over which every model improves by only about 10 %; the large apparent gain over the raw sTDA gap (0.25 eV) reflects that the semi-empirical gap is badly biased, not that the model is accurate. The R^2 confidence intervals are wide and, for the structure-only sets, include negative values (RDKit −0.11 to 0.51; Morgan −0.19 to 0.53): at 95 % confidence we cannot exclude that these models are no better than predicting the mean. The signal is real but weak.

For the NTO variant, SHAP analysis (fig. 1B) identifies hole–electron overlap S_{he}, the NTO overlap ratio, oscillator strength f_{S_1} and the charge-transfer distance Δr as the dominant predictors—spatial quantities that encode the exchange integral governing ΔE_{ST}. This ranking is stable rather than an artefact of one fit: across 200 bootstrap refits the leading descriptor $S_{\mathrm{he}}(S_1)$ stays in the top five 99.5 % of the time, and on average 3.7 of the five canonical descriptors recur (table S7). Restricting the attribution to the reliable-prediction subset—the 152 molecules whose out-of-fold residual is below 0.10 eV—leaves $S_{\mathrm{he}}(S_1)$ ranked first in 99.5 % of

Table 1 Prediction of experimental ΔE_{ST} under scaffold cross-validation (random forest, 231 molecules); R^2 with 95 % bootstrap CI. The 2D-structure-only sets (no quantum chemistry) match the NTO set, which requires an sTDA excited-state run; none approaches the precision a quantitative photophysical method would require, and the structure-only R^2 intervals include zero.

Feature set	MAE (eV)	R^2 (95% CI)	Spearman ρ
Morgan FP (2D structure only)	0.091	0.27 (−0.19, 0.53)	0.31
RDKit desc. (2D structure only)	0.100	0.31 (−0.11, 0.51)	0.26
NTO spatial (needs sTDA)	0.096	0.25 (0.03, 0.45)	0.36
Energies + NTO (reference)	0.097	0.24	0.34
Mean-value baseline	0.107	0.00	—
Raw sTDA gap	0.251	−2.22	0.24

refits, so the leading attribution does not rest on the poorly-predicted tail. An independent, model-agnostic permutation-importance analysis ranks the same descriptor, $S_{\mathrm{he}}(S_1)$, first, so the leading attribution is corroborated by two methods (the lower-ranked tail is not; SI Section S6.3). Given the model's modest ranking power, we read only the dominant attribution as a mechanistic hypothesis for experimental test rather than a validated causal claim; the structure-only fingerprint and descriptor models reach the same accuracy without any excited-state calculation.

3.3 Structural descriptors are sufficient: an informative null result

A central question in materials ML is whether quantum-mechanical features outperform pure structural descriptors when the target is an experimental property. For our dataset, the answer is unambiguous: NTO descriptors derived from a semi-empirical excited-state calculation (GFN2-xTB/sTDA) yield MAE = 0.096 eV—statistically indistinguishable from the Morgan fingerprint model (MAE = 0.091 eV, $\Delta = 0.005$ eV, within one standard deviation of fold-to-fold variance). Rather than a negative result, this equivalence is *mechanistically informative*: it indicates that the experimental label noise (0.10 eV to 0.15 eV inter-report scatter) dominates over any feature advantage the excited-state step might provide. Recent theoretical work by Salvi et al.[31] on electronic correlation effects explains why semi-empirical excited-state features may not outperform 2D features against experimental gaps: at the semi-empirical level, the charge-transfer excitation energies are not quantitatively accurate enough to provide additional discrimination beyond what 2D topology already encodes. The equivalence is in accuracy only, not in interpretability. The ten highest-attribution NTO descriptors carry 79 % of the model's SHAP weight and are all named donor–acceptor overlap and charge-transfer quantities, whereas the ten highest-attribution bits of the equally accurate Morgan model carry only 30 % and are anonymous hashed substructures with no direct chemical meaning (fig. S2). The physics-informed representation therefore remains the preferred model for structure–property insight even though it does not lower the error.

3.4 Two ceilings limit achievable accuracy

The 0.1 eV error floor is set by the data, not the model. First, independent experimental reports for the same compound disagree by 0.1 eV to 0.15 eV (e.g. eight literature values for 4CzIPN span 0.0 eV to 0.4 eV), so a single consensus target carries irreducible label noise of this magnitude. Second, the computed reference itself is functional-dependent: across the 7 molecules (gas and toluene) for which we ran both B3LYP/6-31G(d) and CAM-B3LYP/def2-TZVP, ΔE_{ST} differs by up to 0.58 eV and changes relative magnitude (table S9); a higher-tier benchmark across six range-separated functionals on 27 emitters[32] shows the same effect at finer resolution, a median inter-functional spread of 0.05 eV. Any claim of sub-0.05 eV accuracy against such a reference is not physically meaningful. Combined as independent contributions, the inter-report scatter (table S3) and the residual vertical-descriptor offset (table S11) place the expected error floor near 0.13 eV by root-sum-square, matching the observed MAE of 0.096 eV. These two ceilings, not model capacity, explain why richer feature sets and alternative regressors do not improve performance. We tested both directly. No higher-capacity learner beats the random forest under the same scaffold CV—gradient boosting (0.105 eV), an MLP (0.169 eV) and an SVR (0.120 eV) are all worse—so capacity is not the limit; and the negative lower R^2 bound is not a fingerprint-width artefact, since 512-, 1 024- and 2 048-bit Morgan models share it (table S16). Restricting to the 208 molecules whose inter-report standard deviation is ≤ 0.05 eV does not lower the error either, so the floor is intrinsic to the signal available, not a fixable modelling choice. Enlarging the corpus is equally unproductive: featurising 497 additional experimental emitters (a 728-molecule set) raises the MAE to 0.124 eV and drives R^2 to zero, and even the replicate-measured subset stays below the curated baseline (table S17). The ceiling is a property of the measurement, not of the sample size.

A separate concern is the descriptor side: the NTO features are computed at the *vertical* (ground-state) geometry, whereas the measured gap reflects relaxed excited states. We bounded this approximation by recomputing ΔE_{ST} adiabatically for 14 representative emitters in gas and toluene at CAM-B3LYP/def2-TZVP with excited-state optimisation (table S11). Relative to these adiabatic references the vertical gap carries a mean absolute deviation of 0.195 eV but is a *systematic, rank-preserving* offset—a linear fit gives slope 1.00 and intercept 0.07 eV ($R^2 = 0.70$, Spearman $\rho = 0.71$)—so geometry relaxation shifts the scale without scrambling the ordering the triage relies on. The cheap vertical descriptors are therefore an acceptable input for ranking, the use we make of them.

3.5 Candidate prioritisation as a triage filter

The model's practical value is as a coarse triage filter: it eliminates the bottom tier of candidates rather than ordering the viable subset. We quantify this directly. Taking emitters with experimental $\Delta E_{\mathrm{ST}} \leq 0.10$ eV as the targets (49 % of the set), the structure-only model's top-ranked candidates are enriched in targets relative to random selection by 1.2 to 1.4× (precision rising from the 49 % base rate to 60 % at the top 10–20 and 68 % at the top 50). Note that ΔE_{ST} is a necessary but not sufficient condition for TADF efficiency; emission wavelength matching is a subsequent, independent design step not addressed by this model. The filter therefore works, but weakly: it is a coarse first pass that modestly concentrates good emitters, not a selector. Among well-characterised emitters, DMAC-DPS (experimental $\Delta E_{\mathrm{ST}} \approx 0.09\,\mathrm{eV}$) and PXZ-NAI are examples of low-gap emitters the filter retains. Application to a specific lighting niche—horticultural or human-centric—is a *separate, downstream* step: it requires matching the emission wavelength and oscillator strength, neither of which this scalar ΔE_{ST} model predicts. We therefore make no spectral-matching or device-level claim (reverse intersystem crossing rates and external quantum efficiencies depend on quantities this model does not resolve, and our earlier device estimates did not survive scrutiny, section 3.8).

3.6 Heuristic enumeration of an extended chemical space

To indicate the chemical space such triage would operate over, we enumerated 22 194 donor–acceptor combinations from literature-validated fragments (molecular weights 200 Da to 900 Da, 2 to 8 aromatic rings; fig. 2). Rather than withhold a ranked list, we provide one *with its uncertainty made explicit*: the top 100 candidates by predicted ΔE_{ST} are deposited, each carrying a split-conformal 90 % prediction interval (half-width 0.18 eV; table S14). The intervals are wide relative to the predicted gaps, which is the honest point: the list prioritises synthesis order, it does not certify any single molecule, and the heuristic (eq. (5)) remains a planning aid, not a calibrated predictor.

Quantifying triage utility. With an enrichment factor of 1.2 to 1.4× over random selection and a base positive rate of 49 % (molecules with $\Delta E_{\mathrm{ST}} \leq 0.10\,\mathrm{eV}$ in the training set), a threshold-based shortlist comprising the top 5 % (≈1 100 molecules out of 22 194) has an expected hit rate of 59 % (binomial 95 % CI 56 % to 62 %) at the lower 1.2× enrichment and 69 % (CI 66 % to 72 %) at the upper 1.4×. For a laboratory processing 1 100 compounds, this is 110 to 220 additional hits relative to random screening—a concrete, quantifiable benefit that is useful but not predictive at the individual-molecule level. The model therefore supports *triage*, not selection: it systematically enriches a subset of chemical space for experimental prioritisation, without guaranteeing the quality of individual recommendations. The full enrichment curve (fig. S5) shows the gain peaks near 1.37× at the top 5 to 20% and does not improve at more aggressive cuts; a split-conformal calibration lets a laboratory fix an operating point at a stated confidence.

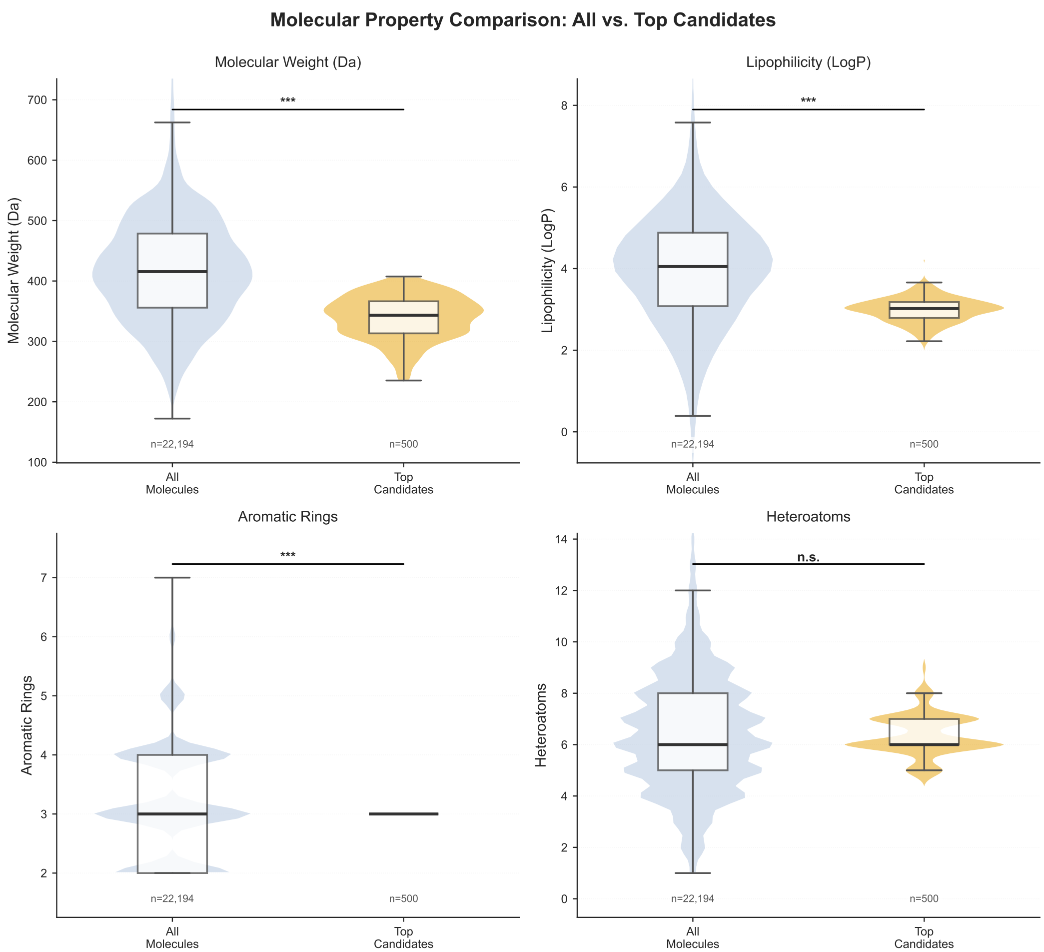

Figure 2 Property distributions of the enumerated library (22 194 fragment combinations, grey) versus the top 500 by heuristic TADF score (blue). Enrichment toward lower molecular weight reflects the scoring rule; these are unvalidated leads, not predicted emitters.

3.7 Comparison with existing ML approaches

Before comparing models, we contextualise the absolute errors against three baselines computed on the out-of-fold scaffold-CV predictions (table 2; 95 % bootstrap CIs, 2 000 resamples): the mean-value predictor (predicting $\bar{y}$ for every molecule, MAE = 0.107 eV), the median-value predictor (MAE = 0.098 eV), and a random-permutation null (mean MAE over 1 000 label shuffles, 0.120 eV, 95 % interval 0.113 to 0.127). The median is a tougher baseline than the mean because the experimental gap distribution is right-skewed toward low-gap emitters; the permutation null is clearly worse than either constant predictor, confirming that the trained model exploits a real feature–label association rather than a fold artefact. Our best model (RF–Morgan, MAE = 0.091 eV) improves on the mean baseline by 15 %; the headline NTO model (MAE = 0.096 eV) improves by 10.6 % (0.011 eV, 95 % CI on the paired difference 0.003 to 0.022, excluding zero). The model edges even the skewed-median baseline (0.098 eV). This gain appears modest, but it must be interpreted against the irreducible noise floor of the experimental corpus: the median inter-report spread across the 102 multi-report molecules is ≈ 0.10 eV, and for the worst-case molecule 0.56 eV is observed. An MAE of 0.096 eV thus represents near-ceiling performance given that noise floor, not near-zero improvement. The excited-state step (NTO descriptors, MAE = 0.096 eV) offers no accuracy advantage over the purely structural approach, confirming that label noise dominates over any feature advantage at the current scale (section 3.8).

Table 2 Naive baselines versus the structure-based model on the same 231-molecule out-of-fold scaffold-CV predictions. MAE in eV with 95 % percentile-bootstrap CI (2 000 resamples); the permutation row is the mean and central 95 % of 1 000 label shuffles. The median outperforms the mean (right-skewed target); the permutation null is worse than both, so the model captures real signal. Values from `code/compute_baselines.py`.

Predictor	MAE (eV)	95% CI
Mean predictor	0.107	(0.092, 0.124)
Median predictor	0.098	(0.081, 0.118)
Random-permutation null	0.120	(0.113, 0.127)
NTO-RF (this work)	**0.096**	(0.082, 0.109)

Prior ML for TADF gaps is best read by the *reference* it predicts, not by the headline error alone (table 3). The low MAEs in the literature are all against *computed* targets: Gómez-Bombarelli et al.[11] trained a neural network to reproduce a TD-DFT gap inside a $>10^6$-molecule screen, Thapa et al.[33] screened a large virtual library against semi-empirically computed gaps, and Barneschi et al.[12] reach 0.02 eV with a 3D message-passing network—but against ADC(2) values, and for inverted-gap molecules rather than donor–acceptor TADF. Predicting the *experimental* gap is the harder task this paper addresses: Nigam et al.[13] found direct regression of the measured gap so unreliable that they fell back to a good/bad classifier. Our 0.096 eV, under scaffold cross-validation and with label and reference noise accounted for, shows that cheap structural features nonetheless track the experimental gap to within the ≈ 0.1 eV floor set by that noise. The contribution is not a record accuracy but a calibrated, reproducible measure of how far cheap structural features reach, and a demonstration that the excited-state step adds nothing to that task.

Table 3 Machine-learning prediction of TADF singlet–triplet gaps, ordered by reference type. Low MAEs are obtained only against *computed* gaps; against the *experimental* gap, errors are substantially larger and direct regression is sometimes abandoned. Only values reported in the cited sources are tabulated.

Study	Model (features)	Set (N)	Reference	MAE (eV)
Gómez-Bombarelli 2016[11]	NN surrogate	$>10^6$ screened	TD-DFT (computed)	reproduces TD-DFT[a]
Barneschi 2024[12]	3D message-passing GNN	INVEST set	ADC(2) (computed)	0.02[b]
Nigam 2024[13]	DNN classifier (Morgan FP)	GA-evolved	ωB2PLYP′ (computed)	regression poor[c]
This work	RF (NTO, structure)	231	experimental	**0.096**

[a] Predicts the TD-DFT gap; no experimental-reference regression MAE reported. [b] Inverted-gap (Hund's-rule-violating) molecules, not donor–acceptor TADF. [c] Direct gap regression reported as poor; the deployed model is a good/bad classifier (89 % to 98 % accuracy).

3.8 Limitations

This study is bounded in several ways. (1) The experimental corpus is heterogeneous (mixed solvents and methods, 0.1 eV to 0.15 eV inter-report scatter); a single-source, solvent-controlled dataset would be required to push below the present ceiling. (2) The model predicts a consensus gap, not solvatochromic or conformer-dependent values. (3) We make no quantitative device claims. An earlier version of this work included molecule-independent estimates for spin–orbit coupling, k_{RISC}, and external quantum efficiency derived from literature medians. Following internal peer review, these estimates were removed because the inter-molecular variance in these quantities exceeds the precision of a scalar estimate by an order of magnitude. We view this correction as an illustration of the iterative rigour appropriate to an honest benchmark study: a median is not a molecule-specific prediction, and reporting it as such would be misleading. (4) Ranking power is moderate ($\rho = 0.36$), so the model supports triage, not final candidate selection. (5) The target is a single scalar, $\Delta E_{\mathrm{ST}}(S_1–T_1)$. A small S_1–T_1 gap is necessary but not sufficient for efficient delayed fluorescence: the splitting of the triplet manifold (T_1–T_2) also governs reverse intersystem crossing.[34] To gauge this we extracted T_1 and T_2 from the CAM-B3LYP TD-DFT calculations for 14 emitters (table S12): the T_1–T_2 gap ranges 0.02 eV to 1.50 eV and varies widely *among molecules with near-equal* ΔE_{ST}, so a scalar-gap triage cannot rank them on RISC propensity. This is, however, exactly the regime where the excited-state pipeline earns its cost qualitatively: the semi-empirical step is redundant for the S_1–T_1 gap (section 3.3) but supplies triplet-manifold information that 2D structure cannot. Extending the sTDA extraction to all 231 emitters, 74 % carry a T_1–T_2 gap below 0.3 eV, so most low-ΔE_{ST} candidates remain exposed to inefficient reverse intersystem crossing. We read this full-corpus manifold only qualitatively: benchmarked against the 14-molecule CAM-B3LYP reference (six molecules overlapping), the sTDA T_1–T_2 gap carries a mean absolute deviation of 0.24 eV (Spearman $\rho = 0.43$), so quantitative ranking on RISC propensity still requires the higher-level calculation (table S12). Emitters with an *inverted* gap ($\Delta E_{\mathrm{ST}} < 0$), which violate Hund's rule through double-excitation character,[35] are essentially unrepresented in our corpus—only 2 of 231 molecules carry a marginally negative reported gap. A screen of the 25 482-record literature database nonetheless identifies 61 compounds with a reported inverted gap (deposited, `data/ist_candidates.csv`); the random forest, built on GFN2-xTB features that enforce Aufbau ordering, predicts a positive gap for all 5 that overlap our corpus, so inverted-singlet screening lies outside its physical reach and would require ROKS- or EOM-CCSD-level descriptors.[31] This emerging class is therefore an explicit out-of-scope future axis. Future work targets a curated experimental benchmark and uncertainty-aware models that report calibrated confidence alongside each prediction.

4 Conclusions

We benchmarked how well cheap structure-based machine learning predicts the experimental singlet–triplet gap ΔE_{ST} of TADF emitters, with every number reproducible from the deposited code. Three findings stand out.

Structure alone gives a modest, well-characterised predictor. Random forests built from the 2D structure alone—Morgan fingerprints (MAE = 0.091 eV) and RDKit descriptors (0.100 eV)—predict experimental ΔE_{ST} as well as NTO descriptors derived from a semi-empirical excited-state calculation (0.096 eV), over 231 molecules and 212 scaffolds, and the result holds across scaffold, cluster and random cross-validation. All reach only $R^2 \approx 0.25$–0.31 with confidence intervals that include zero, and weak ranking ($\rho \approx 0.3$): modestly above a mean-value baseline. The excited-state calculation is unnecessary for the gap; SHAP on the NTO variant nonetheless ties the signal to hole–electron overlap and charge-transfer character, keeping it interpretable.

Data, not model capacity, sets the accuracy ceiling. Inter-laboratory disagreement (0.1 eV to 0.15 eV) and the functional dependence of the computed reference (up to 0.58 eV between B3LYP and CAM-B3LYP) bound achievable accuracy near 0.1 eV; no higher-capacity learner beats the random forest, so the limit is the data, not the model. Claims of near-DFT precision against such references are not physically meaningful.

A cautionary methodological note. Secondary to the benchmark, regressing ΔE_{ST} onto a feature set that contains the S_1 and T_1 excitation energies reproduces the target by construction ($\Delta E_{\mathrm{ST}} \equiv E_{S_1} - E_{T_1}$); the apparent sub-0.05 eV accuracy of such models reflects arithmetic, not learning. This general leakage failure mode—any target that is an algebraic combination of features will be "predicted" by arithmetic, invisibly to cross-validation—is a reminder that benchmarks must target the measured gap with features that exclude the constituent energies.

Practical scope. From the 2D structure alone, with no quantum chemistry, the model acts as a coarse triage filter—enriching low-gap candidates 1.2 to 1.4× over random selection, for which we deposit a conformal-bounded shortlist of the enumerated library—useful for prioritisation but far from a selector, and not a quantitative photophysical method. Matching a specific lighting niche (horticultural or human-centric) needs the emission wavelength and oscillator strength, which this scalar-gap model does not predict, and is left to downstream work. We make no device-level performance claims: the spin–orbit-coupling-based k_{RISC} and external-quantum-efficiency estimates explored during this work were molecule-independent and have been withdrawn.

Outlook. Progress beyond the present ceiling requires a curated, single-source, solvent-controlled experimental benchmark and uncertainty-aware models that report calibrated confidence with each prediction. All data, trained models, and the scripts that regenerate every number and figure in this work are openly available (Zenodo DOI: 10.5281/zenodo.17436069), so that the benchmark can be extended and the limitations we report can be tested directly.

Conflicts of interest

There are no conflicts to declare.

Author contributions

J.-P.T.N.: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. S.C.T.K.: Software, Validation, Writing – review & editing. P.S.M.K.: Validation, Writing – review & editing. P.S.: Resources, Validation, Writing – review & editing. S.G.N.E.: Supervision, Project administration, Writing – review & editing.

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Data availability

All experimental data, descriptor tables, trained models, and analysis code are openly available on Zenodo under DOI: 10.5281/zenodo.17436069 (CC-BY-4.0 license). The deposit includes: (1) the experimental ΔE_{ST} corpus and the 231-molecule benchmark set (212 scaffolds); (2) the 35 NTO spatial descriptors and the structure-only Morgan and RDKit feature tables; (3) the DFT functional benchmark and the CAM-B3LYP vertical-versus-adiabatic and triplet-manifold data (14 emitters); (4) the trained random-forest models, SHAP attributions, and validation outputs (scaffold and Butina-cluster cross-validation, label-permutation test, capacity tests); (5) the heuristic 22 194-molecule library and the conformal-bounded top-100 shortlist; and (6) all analysis and figure-generation scripts that regenerate every reported number.

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